D07 · Order Management System (OMS/ATP/CTP)

# 02 · ATP & CTP: real-time availability & promise management

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

Available-to-Promise (ATP) calculates what can be committed to a new order from stock and confirmed supply; Capable-to-Promise (CTP) extends the check to whether production or replenishment capacity can make it in time.

## 2 WHY IT MATTERS

A promise based on stale or partial inventory data is a future service failure. **ATP that cannot see in-transit stock or plant inventory under-promises and loses sales; ATP that ignores reservations or quality holds over-promises and breaks trust.** The quality of the promise depends on data latency and completeness more than on the sophistication of the algorithm.

## 3 HOW IT IS DONE

1. **Define the supply sources ATP may use.** Stock by node, in-transit with reliable ETA, confirmed production, minus reservations and holds.
2. **Reduce data latency.** Event-driven or frequent updates instead of overnight batches.
3. **Add CTP where products are made to order.** Link the promise to the capacity plan and production lead times.
4. **Measure promise accuracy.** Share of ATP promises fulfilled as committed, reviewed with root causes.

### IN PRACTICE

**The ATP is only as good as the data feeding it — data quality is the prerequisite**

An ATP with inventory data updated 6 hours ago (by overnight batch process) cannot commit the inventory from the last 6 hours of transactions. Real-time ATP requires real-time data feeds.

**Multi-node ATP requires all inventory sources to be integrated**

An ATP that doesn't see in-transit, store, or production inventory is a partial ATP that underestimates real availability and generates order rejections or unnecessarily long lead time promises.

**ATP cannot commit the same inventory to two channels simultaneously**

The golden rule of ATP: one inventory unit can be committed to only one order at a time. An ATP that allows "overselling" (committing more inventory than available) generates delivery failures that are extremely costly to manage.

**CTP requires integration of ATP with the production capacity plan**

For CTP to work, the OMS must be able to query available production capacity and production lead time for each SKU in real time — requiring integration between the OMS and the APS (Advanced Planning System) or the ERP's MRP.

### ILLUSTRATIVE CASE

*Illustrative case built from typical industry values — not data from a specific company.*

**Illustrative case: Real-time ATP implementation — electronics manufacturer, 3 inventory nodes** — Manufacturer has inventory in 3 nodes (plant, national DC, in-transit) but ATP only queries the national DC.

| Supply source | Current ATP status | Gap impact |
|---|---|---|
| National DC inventory | OK | Visible in current ATP · System baseline |
| Plant inventory (finished goods ready to ship) | HIGH | Not visible in ATP · Orders rejected for "out of stock" when 840 units available at plant |

| Supply source | Current ATP status | Gap impact |
|---|---|---|
| In-transit inventory (trucks with known ETA) | HIGH | Not visible in ATP · $2.4M of in-transit product not considered for delivery commitments |
| Scheduled production orders (in process) | MEDIUM | Partially visible · Only current-day production |

**Result:** Post multi-node ATP implementation: Customer fill rate improved from 88.4% to 94.2% with no inventory changes — the inventory existed but the ATP couldn't see it. 840 units of "ghost inventory" at plant released for sales. $2.4M of in-transit inventory included in ATP logic for future delivery promises.

#### HOW IT IS MEASURED

**ATP Accuracy % (delivery promises generated by ATP that are fulfilled)**

ATP Accuracy %(Orders with fulfilled ATP promise / Total orders with ATP promise generated) × 100

*Benchmark: >97% of ATP promises fulfilled · ATP accuracy <90% indicates incorrect assumptions in the ATP calculation Low ATP accuracy is the clearest indicator that the ATP system has incorrect assumptions — inventory the ATP considers "available" isn't in practice, due to unrecorded reservations or quarantine inventory not reflected.*

**ATP Response Time (time for the system to calculate ATP for an order)**

ATP Response TimeTime from ATP request to system response (milliseconds or seconds)

*Benchmark: <2 seconds for real-time e-commerce ATP · <30 seconds for complex multi-node ATP An ATP that takes >10 seconds to respond cannot be used in an e-commerce environment where the customer expects near-instant availability confirmation.*

#### WHAT YOU WOULD USE

**ATP & CTP**

SAP S/4HANA — aATP (Advanced ATP) — Module: Multi-Node ATP + CTP + AllocationSAP S/4HANA's advanced ATP module queries inventory across multiple sources (warehouses, plants, in-transit, scheduled production) and executes CTP integrating the capacity plan.

Manhattan Active Omni — Real-Time ATP — Module: Omnichannel Inventory AvailabilityManhattan Associates real-time ATP system for omnichannel environments with multiple inventory nodes and multiple sales channels.

## 4 SO WHAT

**Promise from complete, current data.** Include every legitimate supply source, remove reservations and holds, add CTP for make-to-order, and measure how often promises are kept.

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